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Comparative Life Cycle Assessment of Aluminium and Carbon Fiber Bike Frames

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Abstract

As the cycling industry seeks sustainable material solutions, understanding the environmental and performance impacts of carbon fiber and aluminium bike frames is essential. This study presents a comparative Life Cycle Assessment (LCA) of aluminium and carbon fiber bicycle frames, examining the cradle-to-grave impacts from raw material extraction through manufacturing, use, and end-of-life (EoL) recycling. The analysis focuses on energy consumption and global warming potential (GWP). The results reveal that aluminium frames, despite their higher recyclability and energy recovery potential, exhibit a net energy consumption of 464.26 MJ and GWP of 102.61 kg CO₂ eq per bike frame. In contrast, carbon fiber frames demonstrate a higher energy demand of 622.63 MJ but a lower GWP of 85.35 kg CO₂ eq per frame, primarily due to reduced emissions during the manufacturing phase. Additionally, the end-of-life phase shows significant trade-offs: while aluminium benefits from well-established recycling processes, carbon fiber recycling technologies remain energy-intensive and technologically underdeveloped. This paper provides stakeholders with insights into the environmental hotspots of both materials and offers guidance for prioritizing sustainability and performance in material selection for bike frames.

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1. Introduction

The bicycle industry is experiencing substantial growth, driven by rising environmental awareness, health consciousness, and supportive government policies. The global bicycle market is projected to grow at a Compound Annual Growth Rate (CAGR) of approximately 6.7 % from 2023 to 2028, reaching an additional \$25.54 billion by 2028 [1]. This surge is influenced by the increasing popularity of cycling as a sustainable and health-focused mode of transport. Governments worldwide are promoting cycling through infrastructure investments and incentives, making it a key component of urban mobility solutions. With cycling viewed as a practical and sustainable transportation choice, manufacturers are increasingly focused on optimizing bicycle design to meet new performance and efficiency expectations. As a result, performance factors such as weight, aerodynamics, rigidity, and precision are becoming increasingly important. These qualities are significantly influenced by the materials used to

build the bicycle.

Currently, aluminium and carbon fiber composites are the most chosen materials for bike frames due to their favorable strength-to-weight ratios and performance. However, to meet the projected surge in bicycle demand in the coming years, it is crucial to carefully examine the environmental impacts associated with the production of these frame materials. While direct literature on the environmental impact and performance optimization of aluminium and carbon fiber in bicycle frames is limited, substantial research exists on the application of these materials in other industries, mainly automotive and aerospace. Due to the high demand for the increased efficiency of airplanes, reducing the weight of the aircraft has been the central focus of the aerospace industry. In [2], LCA for an aircraft wing skin panel manufactured with six different material options, aiming to estimate the environmental impact of these material variants throughout the component's life is conducted and compared with commonly used aluminium alloy. The results highlight that CO₂eq emission over the entire life cycle of aluminium resulted 6700 times higher than the manufacturing and disposal stages, while it increased 720 times for the CFPA6 Prepreg. The results also showcase that

the impacts of the use phase dominate the total impact of entire life cycle. In the automotive industry, Pervaiz et al. [3] illustrated that carbon fiber is two times stronger and 30% lighter in weight as compared to glass which is being used for automotive applications. The cost of carbon fiber makes it a limiting factor to be used for automobile applications and is used only for high-end applications such as luxury and sports cars. They have also focused on the use of recycled carbon fiber which can minimize costs and the harmful hazards to the environment, thus minimizing the production cost. Meng et al. [4] described the effective method to reduce energy consumption using lightweight vehicle for the purpose of the transport sector and provided a detailed study of recycled carbon fiber which can be implemented for various automobile applications.

Beyond the manufacturing stage, the end-of-life (EoL) disposal of carbon fiber composites is also a critical consideration. Unlike aluminium, which can be more readily recycled, carbon fiber requires specialized disposal methods. Virgin carbon fiber is of high cost, which limits its use in bike industry although recycled carbon fiber can be cost-effective but its discontinuous form with random orientation can create difficulties in terms of handling and processing that has restricted the penetration of it in the consumer market up to now. Advanced composites often use a thermosetting polymer matrix, like epoxy, which poses challenges for traditional recycling processes and therefore has restricted recyclability [5]. Incineration and landfill are the traditional approaches for the disposal of carbon fiber components. However, these methods cause serious risk to the environment by generating emissions and ash which does not make it an preferable option. Recycling has been recognized as a desirable end-of-life option to deal with carbon fiber wastes as recycling has the potential to recover value from the waste materials. Fanran meng et al. [6] in his paper conducts a LCA to evaluate the primary energy demand (PED) and Global warming potential (GWP) of carbon fiber composite recycling technologies by comparing conventional landfill, incineration, and the most commercially/near-commercially viable recovery technologies (mechanical recycling, fluidized bed recycling, pyrolysis recycling, chemical recycling) to identify relative environmental performance. While a substantial amount of research exists on the environmental impacts of aluminium and carbon fiber, studies specifically focused on their use and recycling technologies within the bicycle industry are very limited. Therefore, it is essential to conduct a comprehensive evaluation of the environmental impact of these materials, covering various aspects of the manufacturing process, including carbon fiber production, resin selection, and composite fabrication methods as well as the end-of-life phase which includes different recycling technologies.

In this study a comprehensive comparative LCA is conducted, covering manufacturing, transportation, usage and end-of-life phases of both aluminium and carbon fiber bike frames, to determine the most environmentally friendly option throughout its entire life cycle. In the end-of-life phase, multiple recycling scenarios are evaluated for carbon fiber,

comparing their respective environmental impacts.

2. Methodology

Life cycle assessment is a systematic approach to evaluate the environmental impact of a product or service from its inception to its disposal. The cradle-to-grave model has been used for this assessment which will focus on the environmental impacts of the entire life cycle of both the products that includes raw material extraction, manufacturing, transportation, use phase, and end-of-life treatment.

The energy consumption data is gathered from various databases such as EcoInvent and GaBi, moreover reliable scientific research papers have been analyzed. The kg CO_2 equivalent will be calculated using the carbon intensity of the electricity grid of the particular country[40].

Carbon Intensity of Electricity Generation	kg CO_2 eq./MJ
China	0.16
Germany	0.11
Japan	0.13
Netherlands	0.07
Taiwan	0.18

Table 1: Carbon Intensity of Electricity Generation [40].

For the ease of this assessment, a single-issue life cycle assessment which focuses on evaluating the environmental impacts of this process concerning a specific environmental issue is done. Hence, the midpoint indicator is chosen as the global warming potential that belongs to the impact category of climate change. The life cycle impact assessment is conducted in a "Individualist" cultural perspective since the data gathering is done considering short time perspectives and most of the energy intensive processes can be avoided using technology in the future. For this study, the CO_2 equivalent emissions generated during the production of each bike frame are used as a basis to calculate the associated health impacts. These emissions contribute to global warming and, through various pathways such as rising temperatures, extreme weather events, and shifts in disease patterns, result in a measurable loss of healthy life years globally. The total CO_2 equivalent emissions of a bike frame are translated into DALY, through a characterization factor of $1, 13 \cdot 10^{-8}$ DALY/kg [41], providing an estimate of the potential health burden associated with the impacts of climate change of producing one bike frame.

2.1. Goal and scope definition

A bike frame is the main structural component of a bicycle, providing support for the rider and connecting the other essential parts. The most used materials for these frames of high-performance bikes are aluminium and carbon fiber. For the purpose of this assessment two models of bike frame were chosen with aluminium and carbon fiber material. The weights of the

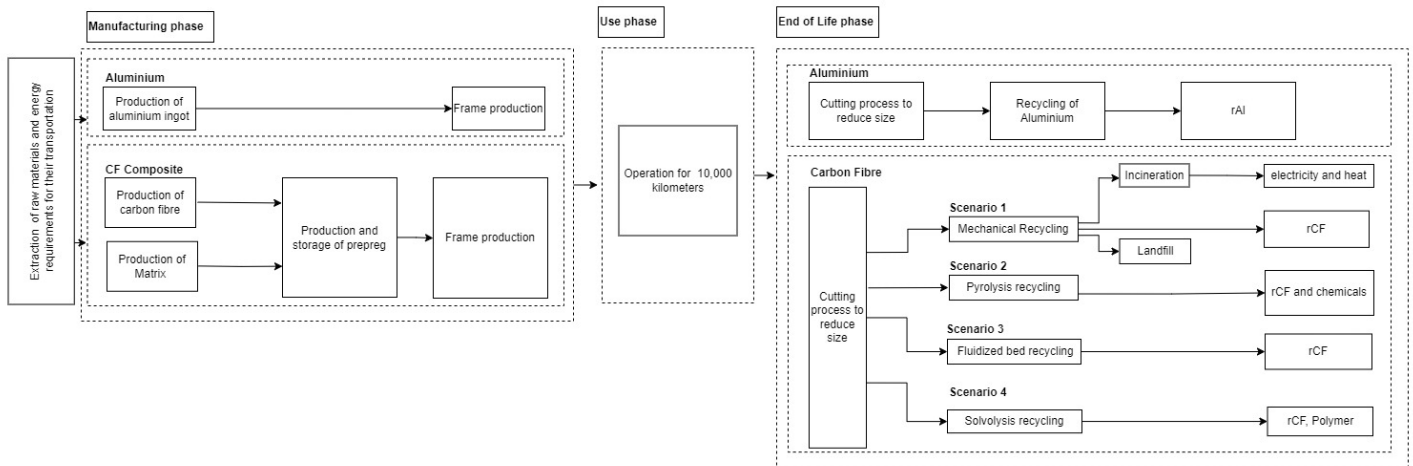


Fig. 1. System boundaries of the LCA.

single bike frames are 2.35 kg and 0.8 kg for the aluminium and carbon fiber respectively, which is the reference flow. The functional unit is defined as “distance covered” and it is assumed that the bike is ridden 10,000 km in its use phase. The LCA takes into consideration the whole life cycle of the products from the raw material extraction till recycling. Energy use, CO₂ eq emission are analyzed for each phase.

The LCA will help manufacturers and consumers make informed decisions about which material offers a more sustainable solution for bike frames and overall ecological footprint of choosing these materials. In addition, the LCA can inform strategies for improving sustainability, identifying key environmental hotspots in the production and disposal processes, such as increasing recycling efficiency or reducing energy use.

The purpose and target audience of the products compared are similar, which makes the results of the assessment rational. The common feature in both the frame sets which is the carbon fork is neglected. Figure 1 represents the phases analyzed and the system boundaries.

2.2. Planned outcomes

The planned outcomes of this LCA are the identification of environmental hot spots in the production and disposal of compared bicycle frames. The carbon fiber frames are expected to have the most energy demand and global warming potential, due to the reason that the recycling rate of carbon fiber is low whereas it is comparatively much higher for aluminium. Moreover, the different methods of recycling technologies for the carbon fiber will be analyzed and the one with the least environmental impacts will be suggested. The analysis of the manufacturing environmental hot spots can be the starting point to propose new advance manufacturing processes that are more environmental friendly.

3. LCA of aluminium Frames

The life cycle of an aluminium bike frame encompasses four primary stages: extraction of raw material, manufacturing, phase of use, and recycling. Each phase contributes differently to the overall environmental impact, with the raw material extraction and manufacturing phases standing out as the most energy-intensive stages due to the high energy demands associated with transforming bauxite ore into aluminium and then into bike frames.

3.1. Raw material extraction

The initial phase of aluminium bike frame production involves mining bauxite, the primary aluminium ore. This extraction process is energy-intensive and often occurs in remote areas, leading to additional energy use for transportation. Primary aluminium production involves extracting alumina from bauxite via the Bayer process, which is used to separate alumina from impurities. This alumina is then processed in the Hall-Héroult electrolysis method that reduces alumina to molten aluminium. The refined aluminium is finally cast into forms like slabs or billets which are called aluminium ingots [8]. For this study, it is considered that the bike frame is produced completely using virgin aluminium extracted in Baise city, China. The data on processes involved in aluminium production were collected from a previous study [9] and are presented in Table 2.

Primary aluminium production

Energy	408.738 MJ / bikeframe
Emissions	65.398 kg CO ₂ eq / bikeframe

Table 2: Energy consumption and CO₂ emissions for aluminium ingots needed for making one bike frame.

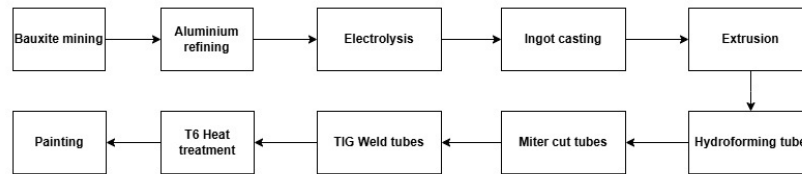


Fig. 2. Manufacturing process of Aluminium bike frame.

3.2. Manufacturing phase

The location of the manufacturing of aluminium bike is considered to be Taichung, Taiwan for this study. The manufacturing of the bike frame begins with the extrusion process to convert into tubes. The detailed steps involved in this phase are represented in Figure 2. The energy consumption and the emission data for this process are based on the energy required to heat the aluminium and generate sufficient force to push it through the extrusion die. Material is lost in every sub process of aluminium extrusion process. Approximately 6% of input material is wasted during extrusion of pre-heated billets in extrusion process and lastly for final cutting process the metal wastage varies from 3.5 to 4% [12]. For the bike frame that is being considered in the study, which is of 2.35 kgs, a total wastage of 10% is assumed, and as a result the total amount of aluminium needed for manufacturing one bike frame will be 2.6 kg of aluminium ingot. Following the extrusion process, the tubes are hydroformed. This process consists in placing the tubes into a mold, high-pressure hydraulic fluid is injected inside, causing the expansion and conforming to the shape of the mold.

Manufacturing process	MJ / bikeframe	kg CO ₂ eq. / bikeframe
Extrusion	19.83 [10]	3.57
Tube Hydroforming	196.56 [13]	35.38
Miter cut	2.85 [14]	0.51
TIG welding	65.11 [15]	11.72
T6 Heat Treatment	1.925	0.34
Painting	12.51	2.25

Table 3: Energy required per aluminium bike frame for the manufacturing processes.

The shaped tubes are cut according to the measurements required for the different parts of the frame, the number of cuts for each bike frame is assumed to be 16. It is assumed that the frames are cut by machining with flood lubrication [14]. Finally, the tubes are connected using TIG welding to realize the final frame, it assumed to perform 11 welds to connect all the shaped tubes together. Each welding is different but making the hypothesis of an average tube diameter to be 50mm, which results in total weld length of 1.7 m. Once the frame is ready, it is heat treated to harden the material. The 6000 series aluminium alloy which is considered for this study are typical heat-treatable aluminium alloys which gain their strength from thermal processing rather than mechanical deformation [16]. This involves two steps— solution heat treating and aging. Solution heat treatment is done by raising the alloy temperature to

about 580°C and holding it there for about one hour. Then the alloy is quenched in water. But for this study, since T6 temper is considered, it is further heat treated at a temperature between 150°C and 200°C, which strengthens the aluminium considerably. Due to unavailability of reliable data for this process, the energy required for this process is calculated based on the specific heat capacity considering the mass of the bikeframe and temperature change. The grid efficiency of 70 percent is incorporated into the calculation. The final operation is painting. The frames are cleaned to remove residues from the manufacturing process, then a primer is applied on the entire frame to enhance adhesion and corrosion resistance. Finally, some layers of high-quality spray color paint is applied.

3.3. Use Phase

The use-phase of the bike frame manufactured is the phase with the least environmental impact as they are human-powered, producing no direct emissions. The use phase of the bike is defined to take place within the Netherlands market. It is also considered that the bike frame is used for 10,000 km and then it is scrapped. This phase is common for both aluminium and carbon fiber frames as the intended function is the same.

3.4. End of Life phase

The recycling phase of an aluminium bike in a LCA examines the environmental impacts and benefits associated with the end-of-life recovery and recycling of the bike's materials. Aluminium is highly recyclable and efficient requiring 95% less energy than producing new aluminium and the process does not alter the properties of the material [14], which significantly reduces the need for raw material extraction and associated energy-intensive processes. The recycling rate for this analysis is assumed to be 95 percent recycling to account for possible losses in the recycling process such as collection and sorting. Hence, the material recovered is calculated for one bike frame and the energy is displaced from the total energy required for manufacturing one bike frame. By recycling, a credit is given for avoided production of primary aluminium which encourages producers to easily facilitate reuse or recycling in the design of products [39]. The recycling of the aluminium bike frames is assumed to occur in the Netherlands, which is also the location of the use phase. In the recycling phase of an aluminium bikes, the transportation for the recycling process is ignored for this assessment due to the variability and unpredictability of logistics involved. Once an aluminium bike reaches the end of its life, it can be sent to various recycling

facilities, each with different capabilities and locations. This decentralized approach complicates the ability to define a specific transportation model, as bikes may travel varying distances based on local recycling infrastructures, collection methods, and market demands for recycled materials. Prior to the recycling phase, the aluminium bike frame is cut into pieces to reduce the size for easy handling. The energy consumption of aluminium recycling processes gathered from the Eco invent database in the regional market for Europe.

Recycling process	MJ / bikeframe	kg CO ₂ eq. / bikeframe
Cutting	0.4	0.028
Recycling	88.4	6.18

Table 4: Energy required per aluminium bike frame for the recycling processes [2].

Material recovered per bike (95% recycling rate)	2.2325 Kg
Material Extraction energy per kg of Aluminium	157.2 MJ
Energy recovered per bike	-350.96 MJ
Kg CO ₂ eq. displaced per bike frame	-24.56

Table 5: Energy recovered and CO₂ emission displaced per bike frame.

3.5. Transportation

The transportation of the materials between different phases are modeled in this study except for the recycling phase. Figure 3 shows the locations and means of transportation for each processes. The distances provided are rough estimates and may differ depending on the specific route taken. The energy consumed for transporting per bike along with the GWP is shown in table 5. The energy required and Kg CO₂ eq. data for different modes of transportation is showed in table 11.

Transportation	Total distance (Km)	MJ / bikeframe	kg CO ₂ eq. / bikeframe
China to Taiwan	2255	3.33	0.312
Taiwan to Netherlands	18390	15.55	1.46

Table 6: Energy consumption and GWP of the Transportation model.

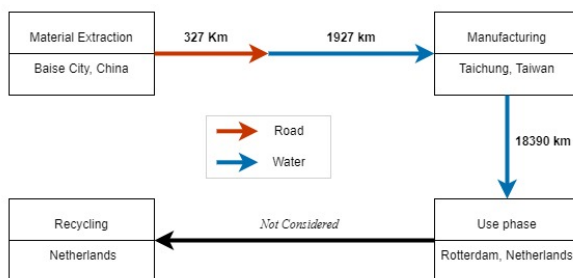


Fig. 3. Location and mean of transportation of each phase

3.6. Aluminium summary

The total energy consumption and kg CO₂ eq. emission for one aluminium bike frame considering the energy recovered and CO₂ emission displaced due to the recycling.

Aluminium bike frame	MJ / bike	kg CO ₂ eq. / bike
Raw material production	408.738	65.398
Bike frame manufacturing	298.799	53.781
End of Life	88.8	6.216
Transportation	18.892	1.782
Gross Total:	815.22	127.17
Energy Recovery and CO ₂ emission displaced per bike	-350.96	-24.56
Net Energy & CO ₂ emission per bike frame	464.26	102.609

Table 7: Total energy consumption and kg CO₂ eq. for one aluminium bike frame.

4. LCA of Carbon Fiber Frame

The life cycle of carbon fiber bike frames can be divided into four different phases: extraction and production of raw materials, manufacturing, phase of use and end of life. All four phases contribute to the overall environmental impact. However, while there is no energy demand in the use phase, the other phases are highly energy intensive.

The Carbon Fiber Reinforced Polymers (CFRP), used for bike frame production, is a fiber-reinforced composite material made from carbon fiber and a thermoset polymer epoxy matrix. It can be manufactured in different ways, but the most common manufacturing method is by using pre-impregnated carbon fibers (prepreg), which are cut, assembled in molds and cured in autoclave [27]. The extraction and production of the prepreg will be analyzed in the raw material production, while the manufacturing covers only the frame production.

For the disposal of carbon fiber, there exists different energy-intensive processes, which will be compared in the EoL.

4.1. Raw Material Production

The production of carbon fibers begins with raw materials such as ammonia and propylene [22]. Through polymerization, these materials are converted into Polyacrylonitrile (PAN) fibers, which then undergo wet spinning and oxidation to form oxidized PAN fibers. [23] Finally, a surface treatment completes the carbon fiber production process [33].

Epoxy resin production begins with core components derived from petrochemical sources, primarily bisphenol-A (BPA) or bisphenol-F (BPF) and epichlorohydrin. These substances undergo polymerization to form a thermosetting polymer that provides the structural matrix, contributing to the resin's overall strength and thermal stability. Various agents and additives are

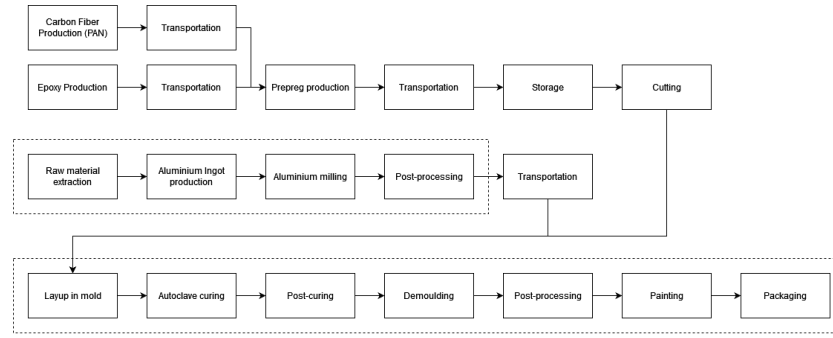


Fig. 4. Manufacturing process of carbon fiber bike frame.

also included, such as hardening agents, plasticizers and accelerators, to create a fully cured, cross-linked epoxy [24] [34].

The prepregging process combines both of these materials into pre-impregnated carbon fibers. The prepreg production begins by weaving the carbon fiber into sheets and coating these with an epoxy resin mixture, ensuring that the resin is evenly distributed to form a consistent matrix around the fibers. Once the resin is applied, the material undergoes a partial curing process, where the resin is partially hardened, to allow the prepreg to be handled and stored without fully curing, maintaining its flexibility for future use. To preserve this state, the prepreg material is rapidly cooled and must be stored at low temperatures, to prevent any further curing and to extend the shelf life up to 12 months [26] [35].

Raw prepreg production	MJ / bike	kg CO ₂ eq. / bike
Carbon fibers	481,92 [23]	64,93 [23]
Epoxy resin	26,21 [23]	3,53 [23]
Prepregging	40,36 [23]	5,43 [23]
Cooling and storage	0.06	0.01

Table 8: Energy consumption per bikeframe of prepreg production.

4.2. Manufacturing

Bike frame manufacturing begins with precision cutting of epoxy resin pre-impregnated carbon fiber sheets, tailored to the frame design for optimal fiber orientation and consistent layer thickness [38].

After cutting, the prepreg sheets are hand-layered into aluminium molds milled from solid blocks for precise dimensions and durability. The aluminium for these molds undergoes a production process similar to that outlined in the aluminium frame LCA for raw material extraction.

Once the prepreg layers are placed in the mold, the assembly is placed in an autoclave, where it undergoes a high-pressure and high-temperature curing process. This process cures the resin and creates a strong bond between the carbon fibers, forming the final composite structure [38].

Following curing, the frame undergoes a series of post-processing steps, including trimming, sanding, and inspection, to remove any excess material and imperfections and refine the frame's surface. These steps ensure that the frame meets precise specifications and achieves a high-quality finish. The

frame is then base-coated to prepare the surface for paint and to enhance adhesion. Finally, a painting process is applied [38].

Carbon bike frame manufacturing	MJ / bike	kg CO ₂ eq. / bike
Prepreg cutting	1.30	0.23
Hand lay-up	14.50 [23]	2.59
Curing in autoclave	16.82 [23]	3.00
Postprocessing	3.00	0.54
Basecoating & painting	10.00	1.78

Table 9: Energy consumption per kilogram for carbon bike frame manufacturing.

4.3. Use Phase

The use-phase of the bike frame manufactured out of carbon fiber is similar to the frame made out of aluminium. This phase has the least environmental impact as the bikes are human-powered, without producing direct emissions. It is considered that the bike frame is used for 10,000 km, before it get scrapped. This phase is common for both aluminium and carbon fiber frames, as the intended function is the same.

4.4. End of Life

Different EoL processes are covered for the carbon fiber bike frame - landfill, incineration, mechanical recycling, pyrolysis recycling, fluidized bed recycling and chemical recycling.

The primary EoL approaches for carbon fiber composites are landfilling and incineration, both resulting in the complete loss of valuable fibers. Landfilling stores non-biodegradable waste, while incineration burns composites for energy recovery, releasing emissions without material reuse [37].

A different EoL approach is mechanical recycling, where the composite is ground into smaller pieces that can be reused as fillers. However, this process significantly reduces fiber length and diminishes the material's mechanical properties [37].

Another EoL option is thermal recycling, including pyrolysis and fluidized bed recycling. Pyrolysis heats materials in the absence of oxygen to decompose the matrix and recover fibers, with slight degradation. Fluidized bed recycling uses high-temperature reactors to separate fibers, resulting in some

loss of strength [37].

The final considered EoL process is chemical recycling (Solvolysis), where chemical solvents dissolve the matrix in a complex and costly process, allowing for the recovery of good quality fibers [37]. As specific data on the distribution of bike frames across various recycling methods is unavailable, the average of all potential EoL scenarios was considered in the conclusion. This approach ensures a balanced assessment by accounting for the different recycling methods, rather than favoring any single process.

Carbon fiber EoL	MJ / kg	kg CO ₂ eq. /kg
Landfill	1.11 [25]	0.14 [25]
Incineration	34 [43]	3.12 [43]
Incineration recovery	-10.51 [44]	1,23 [45]
Mechanical recycling	2.03 [42]	1.80 [25]
Pyrolysis recycling	37.36 [44]	5.4 [44]
Fluidized bed recycling	10.25 [25]	1.56 [25]
Solvolysis	38.39 [25]	2,86 [25]

Table 10: Energy consumption and GWP per kilogram of carbon fiber EoL.

4.5. Transportation

Transportation of materials and products was assumed to follow an optimal approach for a bike company manufacturing in Taichung, Taiwan. The locations and means of transportation for each process are shown in Figure 5. The distances are approximate estimates and may vary depending on the actual route. For each mode of transportation (road, rail, or water), a single representative scenario was selected. All modes involve shipping containers carrying heavy loads, and the well-to-wheel (WTW) consumption was taken into account.

Transportation type	MJ / tkm	g CO ₂ eq. / tkm
Trucks, no trailer	1.8	166.9
Long trains, electric	0.08	10.8
0-999 TEU cargo ships	0.36	34

Table 11: Consumption and emissions of each type of transportation [36]

Transported material	MJ / bike	g CO ₂ eq. / bike
Epoxy ingredients	0.14	10.95
Fiber	0.14	12.80
Prepreg	0.78	73.53
Aluminium (mold)	0.17	16.34
Paint & base coat	0.10	9.25
Frame	6.79	517.84
Total energy consumption	6,79	640.7

Table 12: Energy consumption and GWP per transport route per bike. [36]

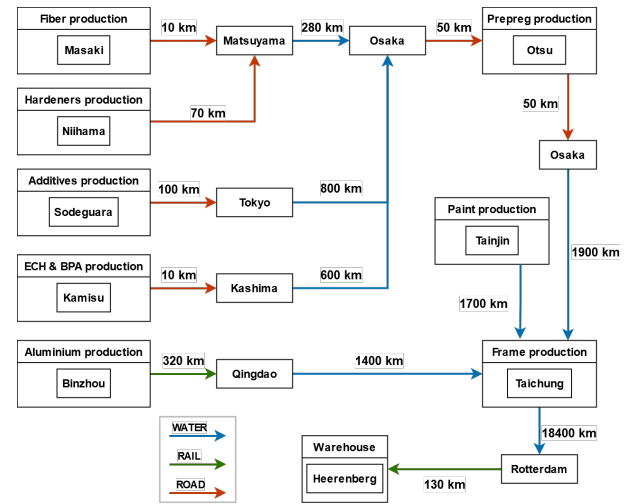


Fig. 5. Location and mean of transportation of each manufacturing process

4.6. Carbon fiber summary

The total energy consumption and CO₂ output for one carbon fiber bike include the EoL value, which is averaged across all previously mentioned methods. This is due to its relatively small impact compared to the overall consumption and the unknown distribution percentages for each method. (Table 13)

Carbon fiber bike frame	MJ / bike	kg CO ₂ eq. / bike
Raw material production	550.40	74.29
Bike frame production	49.88	8.5
End of Life	15.02	1.93
Transportation	6.79	0.64
Total:	622.63	85.35

Table 13: Total energy consumption for one carbon fiber bike frame.

5. Result

The objective of this analysis is to compare the environmental impact of aluminium and carbon fiber bike frames. The comparison focuses on energy consumption (measured in MJ per bike) and global warming potential (measured in kg CO₂ per bike frame). These impacts are assessed across key life cycle stages, including raw material extraction, manufacturing, end of life, as well as transportation.

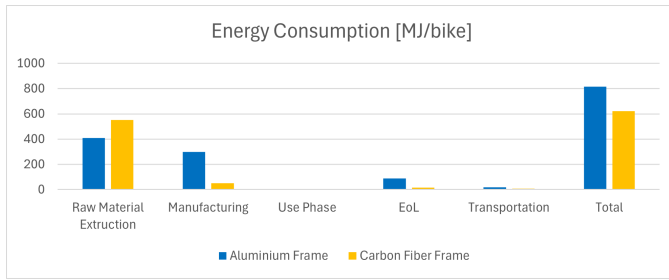


Fig. 6. Energy Consumption Aluminium vs Carbon Fiber Frame. (Does not include the energy recovered from recycling)

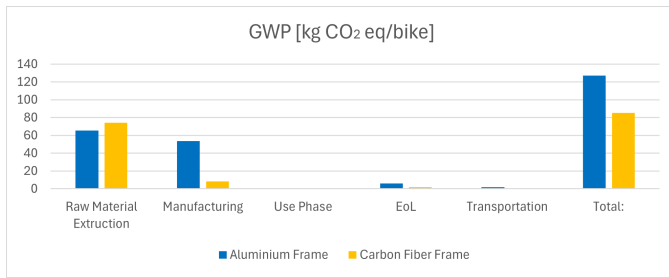


Fig. 7. CO₂ Emissions Aluminium vs Carbon Fiber Frame

The results, visualized in Figures 6 and 7, show that the raw material extraction phase has by far the most significant impact on both energy consumption and GWP for aluminium and carbon fiber frames. For aluminium, the raw material extraction contributes 50.14 % to the total energy consumption, whereas for carbon fiber, it is 89.91%. In terms of GWP, raw material extraction also dominates, with aluminium frames producing 51.42% of overall CO₂ emissions and carbon fiber frames generating 88.17% during this phase.

Out of all the phases of the life cycle of aluminium bike frame, manufacturing phase consumes 36.65 % of the total energy. While for the carbon fiber bike frame the figure is only 8.14 %. The end of life phase for both the bike frames follows a similar outcome. The Aluminium bike frames consume 10.89 % of the total energy while for the carbon fiber frames it is only 1.95 %.

The use phase does not contribute to both energy consumption and emission because the impacts in this phase are determined by the user and are uniform across both materials.

For the Carbon Fiber bike frame, the EoL scenario, visualized in Figure 8, analysis reveals distinct trade-offs. Landfilling has the lowest energy consumption and GWP due to minimal processing, while pyrolysis and chemical recycling (Solvoly-sis) shows the highest impacts, with an increase of approximately 36 - 37 MJ in energy use and 1.4-2.7 kg CO₂ in GWP compared to landfill. Landfilling is currently the most widely used method for CFRP recycling. In most recycling methods, except for solvolysis — which is still very expensive and re-

lies on environmentally harmful substances — the fibers are not recovered with their full material properties. Therefore, it cannot be assumed that PAN fibers can be replaced with recycled fibers. Consequently, no energy use or CO₂ eq. emissions credit can be allocated for this process. Although landfill is the least impactful option currently, it is not aligned with the principles of a circular economy. Recycling processes, while more energy-intensive, hold long-term potential if technological improvements can lower their environmental impacts.

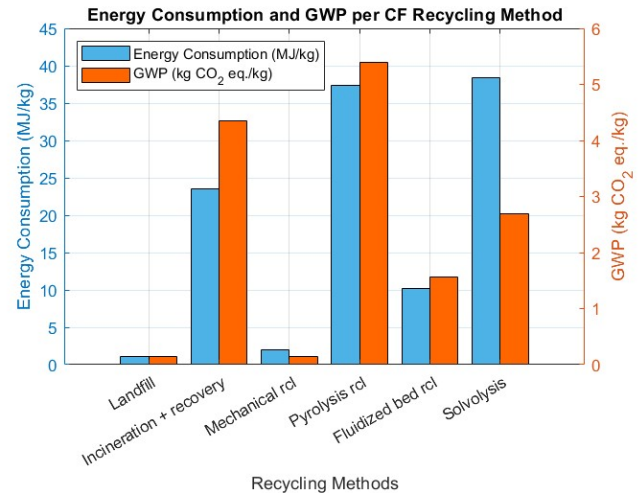


Fig. 8. Energy Consumption and GWP per CF Recycling Method.

Figure 9 and Figure 10 graphs compare the net energy consumption and kg CO₂ equivalent emissions of aluminium and carbon fiber bike frames. In terms of energy consumption, carbon fiber bike frames require significantly more energy compared to aluminium, although energy consumption of individual processes of aluminium bike frame was higher (Figure 6) than carbon fiber. This is due to the fact that the energy of the material extraction is being recovered through the high recycling rate of aluminium which contributes to the overall reduction in the net energy consumption. Whereas, in terms of kg CO₂ eq emissions the net emission of aluminium bike frame is 40% higher than the carbon fiber bike frame. This is because, although material extraction phase of both materials accounts for equal amount of emissions, there is a significant difference in the emission level in the manufacturing phase. The kg CO₂ eq. emission per aluminium bike frame in the manufacturing phase is 32.34 % more than that of carbon fiber bike frames.

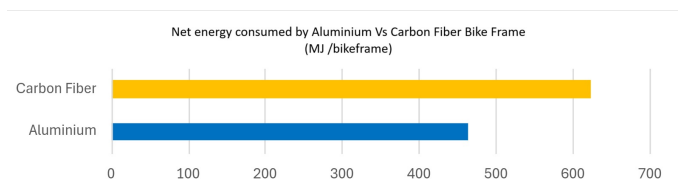


Fig. 9. Total energy consumed by Aluminium Vs Carbon fiber bikeframes

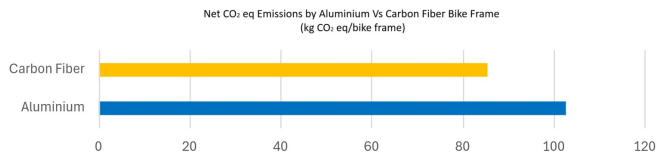


Fig. 10. Total GWP by Aluminium Vs Carbon fiber bikeframes

End-of-life management presents clear opportunities for both materials. For aluminium, maximizing the use of recycled material is key, as recycling significantly reduces environmental impacts. For carbon fiber composites, while landfill currently has the lowest impacts, recycling technologies require further development. Processes like pyrolysis and chemical recycling must become more energy-efficient to surpass landfill as a viable long-term solution. Investing in research and innovation for carbon fiber recycling could support the transition to a circular economy.

It is possible to evaluate the final impacts using the Disability-Adjusted Life Year (DALY) that is a metric used to measure the burden of disease caused by environmental or social factors. This paper focuses only on the carbon dioxide equivalent emissions, which contributes to global warming and indirectly affects human health by increasing the prevalence of diseases and heat-related mortality. The life cycle of an aluminium frame accounts for $1.44 \cdot 10^{-6}$ DALY while a carbon fiber frame contributes to $9.65 \cdot 10^{-7}$ DALY.

6. Conclusion

This study provides a comprehensive comparison of the environmental impacts of aluminium and carbon fiber bike frames using Life Cycle Assessment. By analyzing energy consumption and global warming potential across key life cycle stages, the results highlight the significant trade-offs between the two materials.

The findings show that raw material extraction is the dominant contributor to both energy consumption and GWP for both materials. For aluminium, the environmental burden of this stage can be significantly reduced by maximizing the use of recycled aluminium. Recycling not only decreases energy requirements but also lowers GWP, making it the most effective strategy for reducing environmental impacts associated with aluminium bike frames. For carbon fiber, the production of PAN fibers and resins is the primary source of its high energy demand and CO₂. Efforts to improve efficiency in these production processes or develop alternative, less impactful materials could lead to substantial environmental benefits.

For EoL management, aluminium benefits from well-established recycling processes that significantly reduce its environmental footprint. In contrast, carbon fiber currently presents a challenge. Landfilling has the lowest environmental impact in the short term; however, it does not align with the principles of a circular economy. Recycling technologies,

such as pyrolysis and chemical, require further research and innovation to reduce their energy demands and emissions, making them a more sustainable alternative in the long term. Overall, aluminium frames are currently the more environmentally favorable option, particularly when recycled material is used. However, carbon fiber frames, despite their higher initial environmental impact, offer weight advantages that could provide indirect benefits in applications where performance is critical. Moving forward, manufacturers and policymakers should prioritize the use of recycled aluminium and invest in the development of efficient recycling technologies for carbon fiber. These efforts will be critical for reducing the environmental impact of bike frame production and contributing to a more sustainable cycling industry.

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